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## UNIT 8 CONSUMPTION AND ASSET PRICES\*

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### 8.0 OBJECTIVES

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After going through this Unit you should be able to explain

- the Fisherian idea of consumption as an outcome of households' intertemporal choices;
- the Life-cycle hypothesis of Modigliani and Brumberg and the Permanent Income hypothesis of Friedman;
- the Random Walk Hypothesis of Hall; and
- the asset price determination through the Consumption Asset Pricing (CAPM) model.

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### 8.1 INTRODUCTION

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In Economics, there exist close links between the current economic variables and their past and future values. As you know from economic growth models, past state often determines the current state, just as the current state determines the future. But there is also a link from future to the present. Future of course is often unknown. However, expectations about future variables sometimes influence the current economic decisions. In other words, the decision making process of an economic agent is often 'inter-temporal' in nature -- involving different periods of time. Consumption and savings are two prime examples of such inter-temporal decision making.

How does a household decide upon how much to consume today and how much to save? Keynes identified current income as the prime determinant of current consumption. In its simplest form, a Keynesian consumption function can be represented by the following linear equation:

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$$C_t = \bar{C} + cY_t, \quad \bar{C} > 0, \quad 0 < c < 1, \quad \dots(8.1)$$

where  $C_t$  is current consumption and  $Y_t$  is current income. Notice that one important feature of this consumption function is that the *average propensity to consume* ( $C_t/Y_t$ ) falls as income rises.

Early empirical studies using cross-sectional household data found evidence in support of decreasing average propensity to consume: it was found that richer households (with higher  $Y_t$ ) indeed consumed a lower fraction of their income. However, in 1940s Simon Kuznets using long run time series data on aggregate consumption and income found that the average propensity to consume remained more or less constant, even though aggregate income increased substantially over the period under consideration. This apparent puzzle led to the development of a number of theories of households' consumption behaviour, all of which focus on the intertemporal nature of consumption expenditure. The following section discusses some of the important theories from this literature.

## 8.2 CONSUMPTION AS INTERTEMPORAL CHOICE

One of the earliest works that depicts households' consumption expenditure as an outcome of households' intertemporal optimization exercise is that of Irving Fisher (1930). The Fisherian idea can be easily explained in terms of a two period model of consumer optimization. Let us assume that each member of a household lives for exactly two periods<sup>1</sup> and let  $C_1$  and  $C_2$  denote his consumption during the first and the second period of his life respectively. The person derives utility from both period's consumption and his preferences are represented by the utility function  $U(C_1, C_2)$ . Let  $Y_1$  and  $Y_2$  be the income of the individual in the two periods respectively. Out of his first period income, the person consumes  $C_1$  and saves the rest ( $S_1 = Y_1 - C_1$ ) to earn an interest income  $(1+r)S_1$  in the next period. In Fisher's model, the only purpose of savings is future consumption. Accordingly, his second period consumption is equal to  $C_2 = Y_2 + (1+r)S_1$ .

By re-arranging the terms in  $C_2 = Y_2 + (1+r)S_1$ , we find that  $\frac{C_2}{1+r} = \frac{Y_2}{1+r} + S_1$ .

Since  $S_1 = Y_1 - C_1$ , we have  $\frac{C_2}{1+r} = \frac{Y_2}{1+r} + (Y_1 - C_1)$  or  $C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$ .

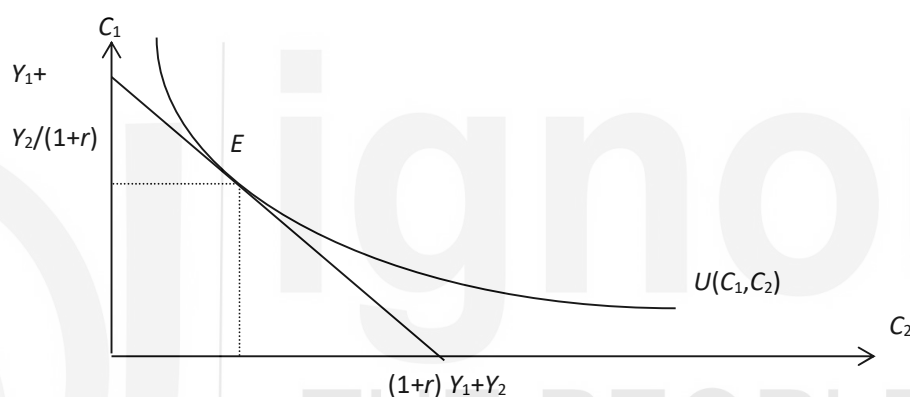
Thus the intertemporal budget constrain is given by

$$C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r} \equiv \hat{Y}(\text{say}) \quad \dots (8.2)$$

<sup>1</sup> The time periods can be broadly defined so that the first time period covers his entire youth and the second one covers his entire old age).

The left hand side of the intertemporal budget constraint denotes the present discounted value of the total consumption expenditure of the person, while the right hand side measures the present discounted value of his total life-time income. The person decides on the levels of  $C_1$  and  $C_2$  by maximizing his utility  $U(C_1, C_2)$  subject to his intertemporal budget constraint. Notice that the optimization exercise of the household defines the current consumption as a function of the present discounted value of life time income ( $\hat{Y}$ ) as well as the rate of interest ( $r$ ) on saving<sup>2</sup>.

In Fig. 8.1 we depict the optimal consumption choices. We measure  $C_2$  on the x-axis and  $C_1$  on the y-axis. The intertemporal budget constraint intersects the x-axis at point  $(1+r)Y_1 + Y_2$  because when  $C_1=0$  in (8.2) we have  $C_2 = (1+r)Y_1 + Y_2$ . Similarly, we find that  $C_1 = Y_1 + \frac{Y_2}{1+r}$  when  $C_2 = 0$ .



**Fig. 8.1: Intertemporal Utility Maximisation**

It is now easy to analyze the impact of a change in current income ( $Y_1$ ) on current consumption ( $C_1$ ). Under the assumption that consumption in both periods are normal goods (i.e., associated with positive income effects), an increase in current income, *ceteris paribus*, increases the life-time income and hence both  $C_1$  and  $C_2$  will increase.

The impact of a change in the interest rate ( $r$ ) on current consumption is more ambiguous. Note that an increase in  $r$  implies a decline in the relative price of future consumption in terms of current consumption.<sup>3</sup> Such a change in the relative price is typically associated with two effects: (i) an *income effect*, which in this case will lead to an increase in consumption in both the periods (since a decline in the price level implies the choice set of the consumer becomes broader), and (ii) a *substitution effect*, which in this case will lead to a fall in current consumption (since a decline in the relative price of future consumption implies that people will substitute current consumption by future consumption).

<sup>2</sup> Strictly speaking, the rate of interest  $r$  is the 'expected' future rate of interest (which is expected to prevail in period 2). We are assuming here that future is certain and known.

<sup>3</sup> This is because for the same amount of current consumption foregone (in the form of savings), one will now get higher amount of future consumption.

The overall effect of an increase in  $r$  on current consumption depends on the relative strength of the two effects. In *ceteris paribus* an increase in  $r$  leads to an increase in  $C_1$  if the income effect dominates the substitution effect; on the other hand, an increase in  $r$  leads to a decrease in  $C_1$  if the substitution effect dominates the income effect. In this context also note that in so far as  $S_1 = Y_1 - C_1$ , an increase in the rate of interest would also imply higher or lower savings depending on the relative strength of the income and the substitution effect.

This Fisherian view of looking at consumption as an outcome of an intertemporal optimization exercise on the part of the households came to play a key role in the development of the subsequent influential theories of consumption, which are: (a) *the life-cycle hypothesis* of Modigliani and Brumberg (1954), and (b) *the permanent income hypothesis* of Friedman (1957). Both these theories attempt to explain the empirically observed discrepancy between the cross-sectional and time series evidence on the relationship between the average propensity to consume and the income level.

### 8.2.1 Life Cycle Hypothesis

To explain the life-cycle and permanent income hypotheses, let us extend the two period model that we have just discussed to a  $T$ -period model where each person lives for  $T$  periods (where  $T \geq 2$ ). The utility function of the representative member of the household is again given by  $U(C_1, C_2, C_3, \dots, C_T)$ . For expositional simplicity let us assume an additive utility function of the form:

$$U(C_1, C_2, C_3, \dots, C_T) = u(C_1) + u(C_2) + \dots + u(C_T) = \sum_{t=1}^T u(C_t). \quad \dots (8.3)$$

The utility function (8.3) is well behaved in the sense that marginal utility is positive (in symbols  $u' > 0$ ) and increases at a decreasing rate (implies that the second derivative is negative,  $u'' < 0$ ). For simplicity let us also assume that the rate of interest is zero<sup>4</sup> so that the intertemporal budget constraint of the household now becomes:

$$\sum_{t=1}^T C_t = A_0 + \sum_{t=1}^T Y_t, \quad \dots (8.4)$$

where  $Y_1, Y_2, \dots, Y_T$  are the incomes of the household in periods 1, 2, ...,  $T$  respectively and  $A_0$  is the amount of initial level of wealth stock of this household. Maximization of utility subject to the intertemporal budget constraint will yield a Lagrangian function of the form:

$$L = \sum_{t=1}^T u(C_t) + \lambda \left( A_0 + \sum_{t=1}^T Y_t - \sum_{t=1}^T C_t \right), \quad \dots (8.5)$$

where  $\lambda$  is the associated Lagrangian multiplier. First order conditions for optimization yield:

<sup>4</sup> Having a positive rate of interest will not have any effect on the subsequent analysis.

$u'(C_t) = \lambda$  for  $t = 1, 2, \dots, T$ , where  $u'(C_t)$  is the marginal utility in period  $t$ .

The above condition implies that  $u'(C_1) = u'(C_2) = \dots = u'(C_T) = \lambda$ , and therefore  $C_1 = C_2 = \dots = C_T$ . Hence from the intertemporal budget constraint (8.4) we obtain

$$C_t = \frac{A_0}{T} + \frac{1}{T} \left( \sum_{t=1}^T Y_t \right) \text{ for all } t. \quad \dots(8.6)$$

The term within the parenthesis in the right hand side of (8.6) is the average value of the total life-time income of the individual. The optimization exercise indicates that consumption in any period is determined the individual's total life-time income. Let us denote the latter by  $\hat{Y}$ . Then consumption at any point of time  $t$  is equal to  $(A_0 + \hat{Y})/T$ . This phenomenon, whereby the individual divides his total life-time resources equally among each period and consumption at different points of time are spread evenly over the entire time horizon is called *consumption smoothing*.

At any particular point of time  $t$ , the actual income of that period  $Y_t$  may exceed or fall short of the average life-time income  $(\hat{Y}/T)$ . Given initial wealth, since current consumption depends only on the average life-time income, therefore any change in current income will have an effect on current consumption *only to the extent that it impacts upon the average life-time income*. To be more precise, suppose at some time period, say  $t = \hat{t}$ , current income rises by an amount  $Z$  for some reason. The corresponding increase in average life-time income is equal to  $Z/T$  and as a result, consumption at period  $t = \hat{t}$  increases only by the amount  $Z/T$ . By the same token, if the current income at some period rises by an amount  $Z$  and the income at some subsequent period falls by the same amount, so that the average life-time income remains unchanged, then current consumption does not change in any period.

Notice that while changes in current income have limited impact on the consumption of that period, current income plays a crucial role in determination of current savings. Recall that savings at any time period  $t$  is defined as  $S_t = Y_t - C_t$ .

By using the result obtained at (8.6) we get

$$S_t = Y_t - \frac{1}{T} (\sum_{t=1}^T Y_t) - \frac{A_0}{T} \quad \dots(8.7)$$

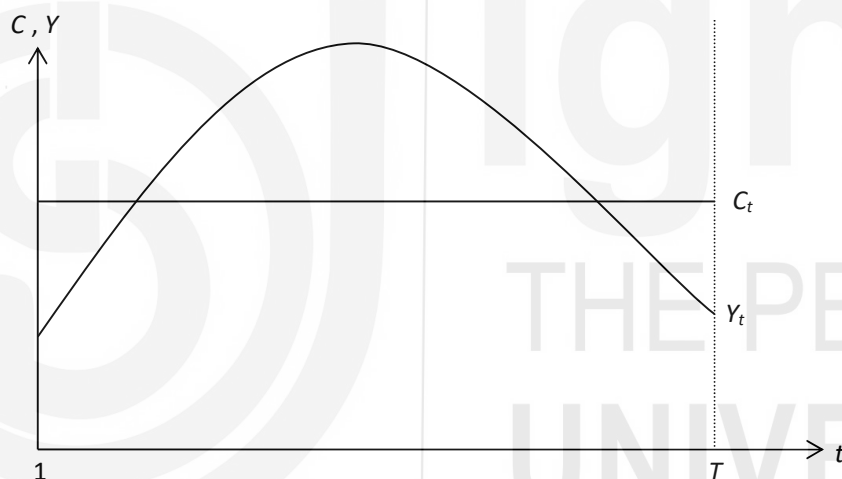
An interpretation of (8.7) is that savings is high when current income is high relative to the average life-time income. It is this idea which forms the basis of Modigliani-Brumberg's life-cycle hypothesis.

According to Modigliani and Brumberg, income varies systematically over the life-time of an individual and savings allow people to smoothen consumption over their life-time, even when incomes in different periods are not equal. In other words, people maintain an even stream of consumption over their life-time

by saving more during high income periods and saving less during the low income periods.

The life-cycle hypothesis postulates that an individual typically has an income stream which is low during the beginning and towards the end of one's life, and high during the middle years of one's life. This is because productivity of a person is typically low during the early and the late years of his life, and productivity is at the peak during the middle years. On the other hand, consumption at every period remains the same at  $C_t = \frac{A_0}{T} + \frac{1}{T} \left( \sum_{i=1}^T Y_i \right)$ . The

typical income and consumption stream over the life-time of the individual is represented by Fig. 8.2. We observe from Fig.8.2, that during the early periods of one's life, an individual dissaves (by running down his initial wealth, and/or by borrowing); he saves during the middle periods of his life and dissaves again during the later years by running down his accumulated wealth.

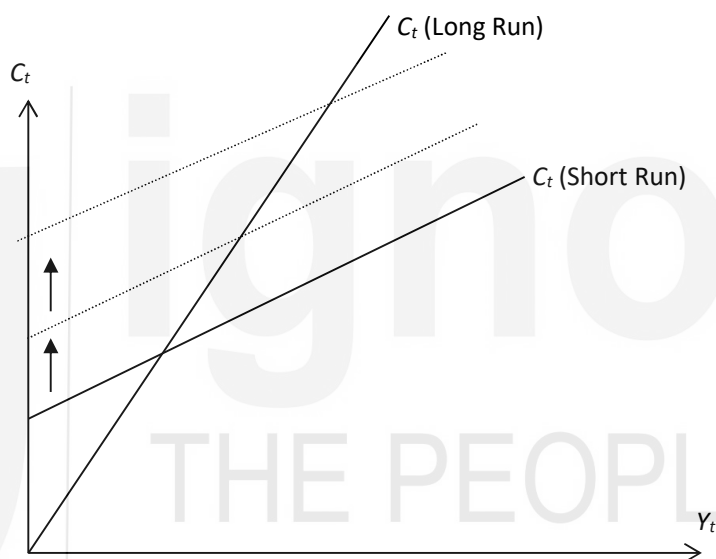


**Fig. 8.2: Consumption Stream over Lifetime**

Given that consumption and savings are postulated to behave in this manner over the life-cycle of a person, it is now easy to see how this hypothesis can explain the apparent disparity between cross-sectional and aggregate time series data on consumption behaviour. When one is looking at the cross-sectional data for a particular point of time, it is likely that in a randomly selected sample of households (which are classified according to income), the high income category will contain a higher-than-average proportion of people belonging to middle phase of their life cycle, and the low income category will contain a higher-than-average proportion of people belonging to either the early or the late phases of their life. Since during the middle phase, people have a higher average propensity to save (i.e., a higher  $S_t/Y_t$  ratio) and correspondingly a lower average propensity to consume (i.e., a lower  $C_t/Y_t$  ratio), the cross-sectional data will

reflect a lower propensity to consume for the high income category compared to the low income category.

At this point it is important to note here that current consumption  $C_t = \frac{A_0}{T} + \frac{1}{T} \left( \sum_{i=1}^T Y_i \right)$  depends not only on the life-time income of the individual, but also on the initial stock of asset  $A_0$ . In the short run (or at any particular point of time) the asset stock remains constant. As a result, if we diagrammatically depict the above relationship between  $C_t$  and  $Y_t$  (with  $C_t$  along the vertical axis and  $Y_t$  along the horizontal axis), then we get an upward sloping line (slope less than unity) with a positive intercept, as shown in Fig. 8.3. However, when we are looking at the time series data where income is



**Fig. 8.3: Short run and Long run Consumption**

increasing over a long period of time, then the stock of asset (which is positively correlated to the level of income) is also increasing. Thus the intercept term on the line will keep on increasing. In other words, the short run  $C_t$  line will keep shifting upward over time.(see Fig. 8.3). Hence if we are examining the long run consumption data to determine the long run relationship between  $C_t$  and  $Y_t$ , we will observe points like **A**, **B**, **C** along successive short run consumption lines. In other words, the long run consumption locus will be steeper, where  $C_t$  is proportional to  $Y_t$ , as shown in Fig. 8.3. Note that along this long run consumption line, the average propensity to consume ( $C_t / Y_t$ ) remains constant.

### **8.2.2 Permanent Income Hypothesis**

Friedman's permanent income hypothesis provides an alternative explanation to the apparent discrepancy between cross-sectional and time series data on

consumption. The permanent income hypothesis is also founded on the Fisherian theory of consumption as an intertemporal choice. However, unlike the life-cycle hypothesis, Friedman does not postulate that income follow a regular pattern over the life cycle of an individual; he instead argues that individuals experience random and temporary changes in their income from time to time. Accordingly, Friedman views the current income in any period ( $Y_t$ ) as consisting of two components: *permanent income* ( $Y_t^P$ ) and *transitory income* ( $Y_t^T$ ). Permanent income is that part of the income which expect to prevail over the long run. Friedman interprets this as the long run average income of the individual; i.e.,

$$Y_t^P = \frac{1}{T} \left( \sum_{t=1}^T Y_t \right).$$

Transitory income is any random deviation from this average, i.e.,  $Y_t^T = Y_t - \frac{1}{T} \left( \sum_{t=1}^T Y_t \right)$ . A positive transitory income implies that the current

income exceeds the permanent income; a negative transitory income implies that the current income is less than the permanent income. Since

$$C_t = \frac{A_0}{T} + \frac{1}{T} \left( \sum_{t=1}^T Y_t \right) = \frac{A_0}{T} + Y_t^P, \text{ i.e., current consumption of the household}$$

depends only on the permanent income, any increase in the transitory part of the current income, which leaves the permanent income unchanged, will have no impact on the level of current consumption.

Let us now see how Friedman's permanent income hypothesis solves the apparent puzzle in the consumption data. According to Friedman's hypothesis, the average propensity to consume ( $C_t / Y_t$ ) depends on the ratio of permanent to current income  $Y_t^P / Y_t$ . Thus when current income temporarily rises above the permanent income the average propensity to consume falls; the opposite happens when current income temporarily falls below the permanent income. Now when we are looking at cross-sectional data of different households at any point of time, typically the high income group will contain some people with a high transitory income, who will have a lower propensity to consume than the average. Similarly the low income group will contain some people with a low transitory income, who will have a higher propensity to consume than the average. As a result, we will observe a falling average propensity to consume as we move from the lower to the higher income group. On the other hand, when we are considering long run time series data, the random fluctuations tend to even out so that any increase in income in the log run reflects a permanent increase in the average income level. Hence in the log run time series data we are likely to observe a constant average propensity to consume.

### Check Your Progress 1

1. Explain the Fisherian idea of consumption as an intertemporal choice. What are the impact of a change in income and a change in the interest rate on current consumption?

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2. What is the puzzle in the consumption data that has been identified by Kuznets? How do the Life-cycle hypothesis and permanent income hypothesis solve this puzzle?

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### 8.3 CONSUMPTION UNDER UNCERTAINTY: RANDOM WALK HYPOTHESIS

In our discussion so far we have assumed that people know their average life-time income (or the permanent part of the income) with certainty. What happens if there is some degree of uncertainty with the permanent income as well? Obviously, in the presence of uncertainty, people's expectation about the future becomes important. We can extend the logic of the permanent income hypothesis to argue that in the presence of uncertainty, individuals maximize their *expected* utility subject to the constraint that the sum of total *expected* consumption cannot exceed the total value of the *expected* life-time income. In other words, the objective function of a consumer is now given by

$$E[U] = E[u(C_1)] + E[u(C_2)] + \dots + E[u(C_T)] = \sum_{t=1}^T E[u(C_t)], \quad \dots (8.8)$$

which he maximizes subject to the constraint

$$\sum_{t=1}^T E(C_t) = A_0 + \sum_{t=1}^T E(Y_t). \quad \dots (8.9)$$

Note that since future is uncertain, the expectation of people about future comes to play an important role, and therefore how people form their expectation also becomes important for the decision making process. You have earlier been introduced to the rational expectation hypothesis in Block-3. According to the

rational expectations theory the outcomes do not differ systematically from what people expected them to be. The extension of the permanent income hypothesis in the presence of uncertainty combined with the assumption of rational expectation led to the theory of ‘random walk consumption’, as expounded by Robert Hall (1978).

Hall argued that if individual’s consumption indeed depended on their expected average life-time income, and if people had rational expectations, then changes in consumption over time will be unpredictable, i.e., consumption will follow a ‘random walk’. The intuition behind this result is simple: if the life-time income is expected to change at some future point of time and people have this information, then they will use this information optimally (under rational expectation) and will therefore immediately adjust their consumption over different time periods (consumption smoothing) so that current consumption would not change at the time when the actual income changes. Current consumption can change only if there are surprises in the life-time income, which were not anticipated. To put it differently, current consumption can only change due to events which are unpredictable and as a result changes in consumption would also be unpredictable.

Theoretically it is easy to see how this hypothesis follows from the above formulation of individual’s maximization problem. As we saw in (8.6) and (8.7) the individual maximizes  $\sum_{t=1}^T E[u(C_t)]$  subject to  $\sum_{t=1}^T E(C_t) = A_0 + \sum_{t=1}^T E(Y_t)$ . Since the individual is taking his decision at time  $t=1$ , under rational expectations, all his expectations are based on the information available at period 1. An optimizing agent will equate his *expected* (as of period 1) marginal utilities in different periods. Now in period 1, consumption in period 1 is a certain event; hence  $E_1[u'(C_1)] = u'(C_1)$ . On the other hand, *expected* (as of period 1) marginal utility at any subsequent period is given by:  $E_1[u'(C_t)]$ ;  $t = 2, 3, \dots, T$ . Thus the optimality condition implies:

$$u'(C_1) = E_1[u'(C_2)] = E_1[u'(C_3)] = \dots = E_1[u'(C_T)] \quad \dots (8.10)$$

In order to explain (8.10) further let us take a quadratic utility function of the

form:  $u(C_t) = C_t - \frac{a}{2} C_t^2$  for  $t = 1, 2, \dots, T$ . Then the marginal utility function is

linear in  $C_t$ , i.e.,  $u'(C_t) = 1 - aC_t$  for all  $t$ . Therefore,

$E_1[u'(C_t)] = E_1[1 - aC_t] = 1 - aE_1(C_t)$ . Using this in the optimality condition we get:

$$1 - aC_1 = 1 - aE_1[C_2] = 1 - aE_1[C_3] = \dots = 1 - aE_1[C_T].$$

On simplification,

$$C_1 = E_1[C_2] = E_1[C_3] = \dots = E_1[C_T] \quad \dots (8.11)$$

The above condition implies that expectation as of period 1 about  $C_2$  equals  $C_1$ . In more general terms if expectations are formed at any period  $t$  about a future period  $t+1$ , then optimality condition requires that expectation as of period  $t$  about  $C_{t+1}$  equals  $C_t$ , i.e.,  $C_t = E_t[C_{t+1}]$ . Since under rational expectations the actual value of a variable can differ from its expected value only by a random term, this would imply

$$C_{t+1} = E_t[C_{t+1}] + e_{t+1} = C_t + e_{t+1} \quad \dots (8.12)$$

where  $e_{t+1}$  is a random term whose expected value is zero. If you look carefully, equation (8.12) says that consumption from period  $t$  to period  $t+1$  would remain unchanged, except for an unpredictable random term. This is precisely the conclusion of the random walk hypothesis. The hypothesis says that changes in consumption over time is unpredictable, because they can only change due to the presence of unpredictable random events.

The random walk hypothesis has important implications from the point of view of policy effectiveness. It implies that any government policy to influence consumption (for example, a tax cut) can work only to the extent that it is not anticipated. For example, if the government announces a tax cut policy today, which will be implemented from next year, then consumers with rational expectations will adjust their consumption today itself, so that consumption would remain unchanged when the tax policy actually becomes operative next year.

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## 8.4 CONSUMPTION AND RISKY ASSETS: CAPITAL-ASSET PRICING MODEL

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While analysing consumption under uncertainty in the previous Section, we had implicitly assumed that the source of uncertainty is the households' income. In other words, we had assumed that there was no uncertainty as far as the asset return is concerned: the return to asset ( $r$ ) was treated as given. (In fact, for simplicity we had assumed that it has a value equal to zero). If the return to asset is non-zero, but the exact return is uncertain then the optimality condition of the individual's utility maximization will include an expected return term as shown below:

$$u'(C_t) = E_t[(1 + r_{t+1})u'(C_{t+1})], \quad \dots (8.13)$$

where  $t$  is the time period when expectations are formed.

We know from the theorem for statistical expectation (operator  $E$ ) for two events that  $A$  and  $B$   $E(AB) = E(A).E(B) + \text{Cov}(AB)$  (where  $\text{Cov}(AB)$  is the covariance between the two events  $A$  and  $B$ ). We can visualize (8.13) as a case of two joint events and write the optimality condition as:

$$u'(C_t) = E_t[(1 + r_{t+1})].E_t[u'(C_{t+1})] + \text{Cov}_t[(1 + r_{t+1}), u'(C_{t+1})] \quad \dots (8.14)$$

Now suppose there are two assets: i) one with a certain return (i.e., risk-free) given by  $\bar{r}_{t+1}$ , and ii) another with an uncertain risky return with an expected return value equal to  $E_t(r_{t+1})$ . For the risky asset, if we apply the condition (8.14) we obtain

$$E_t[(1 + r_{t+1})] = \frac{u'(C_t) - \text{Cov}_t[(1 + r_{t+1}), u'(C_{t+1})]}{E_t[u'(C_{t+1})]},$$

$$\text{i.e., } 1 + E_t[r_{t+1}] = \frac{u'(C_t) - \text{Cov}_t[(1 + r_{t+1}), u'(C_{t+1})]}{E_t[u'(C_{t+1})]}. \quad \dots (8.15)$$

For the risk-free asset, the return is certain and therefore is uncorrelated to  $C_{t+1}$ . In other words, for the risk-free asset  $\text{Cov}_t[(1 + \bar{r}_{t+1}), u'(C_{t+1})] = 0$ . Hence for the risk-free asset, the condition (8.14) implies that

$$1 + \bar{r}_{t+1} = \frac{u'(C_t)}{E_t[u'(C_{t+1})]}. \quad \dots (8.16)$$

Comparing (8.15) and (8.16) we get:

$$E_t[r_{t+1}] = \bar{r}_{t+1} - \frac{\text{Cov}_t[(1 + r_{t+1}), u'(C_{t+1})]}{E_t[u'(C_{t+1})]}. \quad \dots (8.17)$$

The condition (8.17) gives us a way of determining the optimal (or equilibrium) expected return of a risky asset. Note that a higher value of  $C_{t+1}$  implies a lower value of  $u'(C_{t+1})$ . Therefore a positive co-variance between  $r_{t+1}$  and  $C_{t+1}$  implies a negative co-variance between  $r_{t+1}$  and  $u'(C_{t+1})$ . Hence (8.17) implies that the higher is the correlation between  $r_{t+1}$  and  $C_{t+1}$ , the greater is the required expected return on a risky asset compared to the return from the risk-free asset. To put it differently, the greater is the covariance between  $r_{t+1}$  and  $C_{t+1}$ , the higher is the premium that an asset must offer relative to the risk-free rate. This model of determination of expected return of risky assets is known as the Consumption Capital Asset Pricing Model.

## Check Your Progress 2

1. Describe the Random Walk Hypothesis of consumption. What is the implication of this hypothesis from the perspective of government policy?

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2. Explain the Consumption Capital Asset Pricing rule for a risky asset.

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## **8.5 LET US SUM UP**

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The present unit viewed consumption in a dynamic set up and introduced you to the problem of intertemporal utility maximization. In this context it discussed the issue of consumption and savings when more than one time periods are considered. The Fisherian idea that consumption is an outcome of household's intertemporal optimization exercise has been explained in the unit.

Empirical studies involving cross-section data find that richer households consume a smaller fraction of their income compared to poorer households. However, studies based on time series data show that the proportion of consumption in income has remained more or less the same although household income has increased manifold. This apparent discrepancy has been attempted to be resolved through life-cycle hypothesis and permanent income hypothesis. According to the life cycle hypothesis the short run consumption function shows a declining average propensity to consume (APC) while the long run consumption function exhibits constant APC due to increase in asset base. On the other hand, the permanent income hypothesis explains the discrepancy in terms of permanent income and transitory income.

In the presence of uncertainty actual consumption behaviour may be difficult to predict. In this situation consumption may follow a random pattern. This theory of consumption, known as the Random Walk Hypothesis, has been explained here. Finally, in the presence of uncertainty the price of the risky asset may follow a specific pattern which has been explained in the discussion on the CAPM model.

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## **8.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES**

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### **Check Your Progress Exercises 1**

1. A change in income will shift the inter-temporal budget constraint. A change in interest rate will affect the slope of the budget constraint. Incorporate the changes in Fig. 8.1 and write your answer.
2. Kuznets found that average propensity to consume remained constant while using aggregate time series data; unlike the Keynesian consumption. In order to explain this difference, two hypotheses are propounded – life cycle hypothesis by Modigliani and permanent income hypothesis by Friedman.

### **Check Your Progress Exercises 2**

1. The random walk hypothesis implies that any government policy to consumption can work only if it is not anticipated.
2. It says that return has to be higher from risky assets. This model is used to analyse market risk premium. See Section 8.4 for further details.

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## UNIT 9 RAMSEY-CASS-KOOPMANS MODEL\*

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### Structure

- 9.0 Objectives
- 9.1 Introduction
- 9.2 Central Planner's Problem
- 9.3 Decentralized Households' Problem
- 9.4 Government in Ramsey-Cass-Koopmans Model and Ricardian Equivalence
- 9.5 Let Us Sum Up
- 9.6 Answer/Hints to Check Your Progress Exercises
- 9.7 Mathematical Appendix

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### 9.0 OBJECTIVES

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After going through this Unit you should be able to

- explain the Ramsey-Cass-Koopmans problem of optimal growth;
- compare the optimal growth problem of the central planner and that of the decentralized perfectly competitive economy;
- explain the role of government in the optimal growth framework and the corresponding concept of Ricardian Equivalence; and
- if you go through the Mathematical Appendix carefully you should be in a position to solve any standard dynamic optimization exercise.

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### 9.1 INTRODUCTION

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In our discussion of household's intertemporal consumption and saving decisions in the previous unit, we considered optimization problems where the time horizon was finite. While this is a valid assumption for an individual, a household (or for that matter, the society as a whole) is generally infinitely lived. In other words, we think of a household consisting of individual members, each of whom live for a finite time period, but new members are born in each time period who are an exact replica of the older members. Thus the household as a dynasty continues to live forever.

An important economic question that is often asked in the context of an infinitely lived household is the following: how should the household allocate its resources between consumption and saving in each time period so that the utility level of each of its members is maximized? In other words, what are the optimal consumption path and optimal saving path for this household? To put it more generally, in an economy consisting of *identical* infinitely lived households, what should be the optimal savings/ accumulation pattern that maximizes the utility of

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each of the households over time? This latter question was first tackled by Frank Ramsey (1928), which was later generalized by Cass (1965) and Koopmans (1965). The model has subsequently been known as the the Ramsey-Cass-Koopmans (RCK) *optimal growth* model.

The Ramsey-Cass-Koopmans optimal growth model essentially involves a dynamic optimization exercise. Here, an economic agent maximizes its objective function defined over a period of time (finite or infinite) subject to some constraints. The constraints are also dynamic in nature in the sense that they relate to values of the variables for different time periods. The technique for solving such dynamic optimization exercises has been specified in the Mathematical Appendix given at the end of this Unit. In the discussion below we will simply state the conditions for optimization (without getting into the underlying technique) and explain the economic intuition behind each of these conditions.

There are two versions of the model: (i) the central planner's problem, and (ii) the decentralized households' problem. You should note that Ramsey introduced his model in 1928 as a central planner's problem in the sense that the planner allocates resources between consumption and saving so as to maximize the utility of the households. Solow model, which is a much simpler growth model compared to Ramsey model, was developed by Robert M. Solow in 1957. Subsequently, the Ramsey model was extended by David Cass and Tjalling Koopmans (separately) in 1965 to include the decentralized households' problem. Keeping in view the similarity of approach in both the versions, they are bracketed together.

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## 9.2 CENTRAL PLANNER'S PROBLEM

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In the original Ramsey formulation, the optimal growth problem was postulated as the problem of a social planner who seeks to maximize the welfare of each household subject to the economy's resource constraint at each point of time. The households are all assumed to be identical in terms of preferences and their welfare is represented by an infinite horizon utility function of the form

$$W = \int_0^{\infty} u(c_t) \exp^{-\rho t} dt$$

where  $c_t$  is the *per capita consumption* in period  $t$ ,

$u(c_t)$  is the associated *instantaneous utility* in period  $t$ , and

$\rho$  is a positive constant term which represents the *subjective discount rate* of the household, or its *rate of time preference*.

Before we proceed further, it is important to explain the presence of the discount factor in the household's welfare function. The positive discount factor reflects the household's preference for present over future. In other words, it implies that a household puts more weight on consumption today vis-à-vis consumption

tomorrow. Note that when  $t=0$  (i.e., the initial time period)  $\exp^{-\rho t} = 1$ . Subsequently when  $t=1$ ,  $\exp^{-\rho t} = \exp^{-\rho}$ ; when  $t=2$ ,  $\exp^{-\rho t} = \exp^{-2\rho}$  and so on, such that  $1 > \exp^{-\rho} > \exp^{-2\rho} \dots\dots\dots$ . Therefore in the infinite-time utility function  $W$ , current utility at period zero is associated with the highest weight (unity), and each subsequent utility term is associated with lower and lower weights.

The social planner maximizes the integral  $W$ , but he is constrained by the fact that at each point of time, the economy's total consumption and total investment cannot exceed its total output.<sup>1</sup> To put it formally,  $C_t + \frac{dK}{dt} = Y_t$ , where  $C_t$  is

**aggregate consumption** in period  $t$ ;  $\frac{dK}{dt}$  is the amount of investment in period  $t$  which augments the capital stock ( $K$ ) of the economy and  $Y_t$  is the total output produced in period  $t$ .

Total population at time  $t$  is given by  $L_t$ , which is assumed to grow at a positive exogenous rate  $n$ :  $\frac{1}{L} \frac{dL}{dt} = n$ .

Output at any point of time is produced using the existing capital and labour at that point of time. Technology is represented by a neoclassical production function given by  $Y_t = F(K_t, L_t)$ . You should note that  $F$  is continuous, concave and exhibits constant returns to scale (CRS). The CRS property of the production function implies that per capita output ( $y$ ) can be written as the function of the capital-labour ratio ( $k$ ) in the following way:

$$y \equiv \frac{Y}{L} = \frac{F(K, L)}{L} = \frac{L F(K/L, 1)}{L} = F\left(\frac{K}{L}, 1\right) \equiv f(k).$$

Moreover, the marginal products of capital and labour can also be written as the following functions of the capital-labour ratio:

$$\frac{\partial F}{\partial K} = f'(k) ; \quad \frac{\partial F}{\partial L} = f(k) - kf'(k).$$

The continuity and concavity properties of  $F(L, K)$  ensure that  $f(k)$  is also continuous and concave. Additionally, we assume that  $f(0) = 0$ ;  $f'(0) = \infty$ ;  $f'(\infty) = 0$ . The first of these assumptions implies that no production is possible with zero capital-labour ratio. The last two assumptions are known as the 'Inada conditions' which state that when the capital-labour ratio tends to zero, the marginal product of capital tends to infinity, while an infinitely high capital-labour ratio implies that the corresponding marginal product of capital approaches zero.

The aggregate resource constraint of the economy is given by  $C_t + \frac{dK}{dt} = Y_t$ .

<sup>1</sup> For simplicity, we rule out international borrowing here.

Let us divide both sides of the above equation by  $L$ . Thus, we can write it as

$$\frac{C_t}{L_t} + \frac{1}{L_t} \frac{dK}{dt} = \frac{Y_t}{L_t}$$

$$\text{i.e., } c_t + \frac{dk}{dt} + nk_t = f(k_t)$$

The above equation denotes the resource constraint faced by the social planner in per capita terms.<sup>2</sup>

The economy starts with a certain given amount of capital stock and population; hence the initial capital-labour ratio is given, denoted by  $k_0$ . At every point of time the existing capital and labour stocks are fully employed. Hence, the capital-labour ratio  $k$  also denotes the per capita capital stock. While referring to  $k$  we shall use these two terms interchangeably.

The complete dynamic optimization problem for the social planner can now be written in terms of the two time dependent variables: (i) per capita consumption ( $c_t$ ), and per capita capital stock ( $k_t$ ). The maximization problem before us is

$$\text{Maximize } W = \int_0^{\infty} u(c_t) \exp^{-\rho t} dt$$

subject to

$$\frac{dk}{dt} = f(k_t) - nk_t - c_t; k_0 \text{ given.}$$

The corresponding Hamiltonian function (H) is given by<sup>3</sup>

$$H = u(c_t) \exp^{-\rho t} + \lambda_t [f(k_t) - nk_t - c_t]$$

In the above equation, we have control variable ( $c_t$ ), state variable ( $k_t$ ) and co-state variable ( $\lambda_t$ ). The first order conditions are obtained by taking partial derivatives of the Hamiltonian with respect to  $c_t$ ,  $\lambda_t$  and  $k_t$  and equating it to zero. Thus, we obtain

$$(ia) \quad u'(c_t) \exp^{-\rho t} = \lambda_t$$

$$(iia) \quad \frac{d\lambda}{dt} = -\lambda_t (f'(k_t) - n)$$

$$(iiia) \quad \frac{dk}{dt} = f(k_t) - nk_t - c_t$$

The transversality condition is given by

$$(iva) \quad \lim_{t \rightarrow \infty} \lambda_t k_t = 0$$

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<sup>2</sup> Note that  $\frac{dk}{dt} = \frac{1}{L} \frac{dK}{dt} - \frac{K}{L} \left( \frac{1}{L} \frac{dL}{dt} \right) = \frac{1}{L} \frac{dK}{dt} - nk$ .

<sup>3</sup> See the Mathematical Appendix for the definitions of the Hamiltonian function, control, state and co-state variables in a dynamic optimization problem.

Notice that  $\lambda_t$  is the shadow price of capital (Please go through the Appendix to this Unit) or the value of capital in utility terms. Hence, condition (ia) states that along the optimal path, the present discounted value of marginal utility from consumption would be equal to the shadow price of capital. The economic significance of this condition would become clear if you note that at any point of time one unit of output can be put to two different uses – one can either consume it or invest it which augments the capital stock. Now optimality condition requires that the returns in terms of utility from these two uses should be equal, which is precisely what condition (ia) states. Conditions (iia) and (iiaa) respectively show how the co-state and the state variable changes over time along the optimal path. The fourth condition, known as the **transversality condition**, implies that as the economy approaches its terminal time (which in this case is infinity), either the value of capital goes to zero (in which case it does not matter for the economy if it leaves a positive capital stock at the end), or, if the value of the capital stock is positive, then the economy uses up all its capital stock (i.e.,  $k$  goes to zero).

Differentiating (ia) with respect to  $t$  and using (iia) to eliminate  $\lambda_t$ , we get the following differential equation:  $\frac{dc}{dt} = \left( \frac{u'(c_t)}{-u''(c_t)} \right) (f'(k_t - n - \rho)$ . Noting that the term  $\frac{-cu''(c)}{u'(c)} \equiv \sigma$  denotes the elasticity of marginal utility with respect to consumption, the above equation can be written as:

$$(va) \quad \frac{dc}{dt} = \left( \frac{c_t}{\sigma} \right) (f'(k_t - n - \rho)$$

Equations (iiaa) and (va) together form a system of differential equations which, along with the transversality condition, determine the movements of per capita consumption and per capita capital stock of the economy along the optimal path. The qualitative characterization of this path is shown in the phase diagram given at Fig. 9.3.

The phase diagram traces the lines/curves along which  $\frac{dc}{dt} = 0$  and  $\frac{dk}{dt} = 0$ . The point of intersection of these two lines is called the **steady state**. A steady state is equivalent to long run equilibrium whereby the values of the variables remains constant over time.

From (va),  $\frac{dc}{dt} = 0$  implies either  $\frac{c_t}{\sigma} = 0$  or  $f'(k_t - n - \rho) = 0$ . In the first case,  $c = 0$ . In the second case,  $f'(k) = n + \rho$ . Thus, the equilibrium will remain on these two lines. Let us consider equation (iiaa). When  $\frac{dk}{dt} = 0$ , we have  $f(k_t) - nk_t - c_t = 0$ . On re-arranging terms and ignoring the subscripts (since the values will remain unchanged under equilibrium conditions), we obtain  $c = f(k) - nk$ . Let us first look into the dynamics of the differential equation

$\frac{dc}{dt} = \left(\frac{c_t}{\sigma}\right)(f'(k_t) - n - \rho)$ . Since it can be represented by the lines  $c = 0$  and  $f'(k) = n + \rho$ , we can describe the dynamic of consumption in Fig. 91.

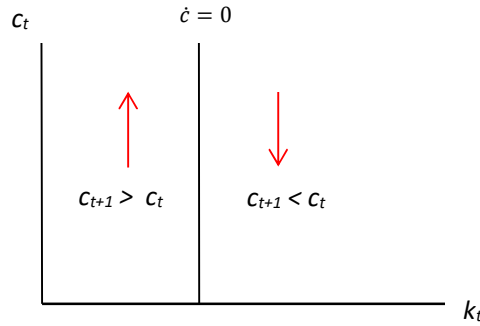
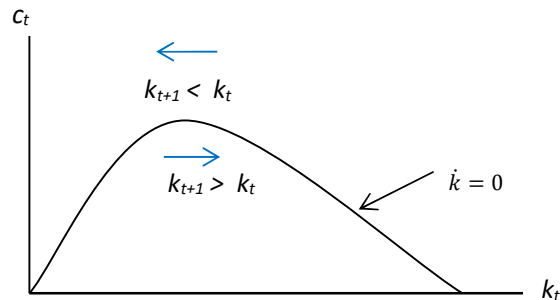


Fig. 9.1: Dynamics of  $c$

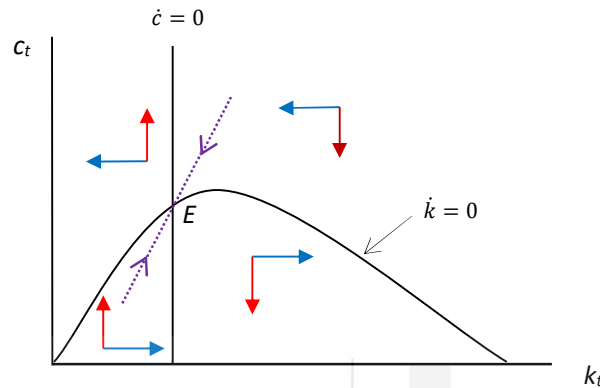
In Fig. 9.1, we take per capita capital stock ( $k$ ) on the x-axis and per capita consumption ( $c$ ) on the y-axis. The vertical line shows the equilibrium condition where there is no change in per capita consumption over time. Therefore, along this line, we have  $\frac{dc}{dt} = \dot{c} = 0$ . To the left of this vertical line, we have  $\dot{c} > 0$ . It implies there will be increase in consumption per capita over time (i.e.,  $c_{t+1} > c_t$ ). We have indicated the direction of such change in per capita consumption by an upward-sloping red arrow. To the right of the vertical line, on the other hand,  $\dot{c} < 0$ , which implies a decrease in per capita consumption over time (thus,  $c_{t+1} < c_t$  in this case). If the economy is operating to the right of the vertical line (indicating per capita consumption at a higher than the equilibrium level), there will be decline in per capita consumption over time. We have indicated the direction of such change in per capita consumption by a downward-sloping red arrow in Fig. 9.1.

Now let us discuss how per capita capital stock evolves over time. When  $\frac{dk}{dt} = \dot{k} = 0$ , we have  $c = f(k) - nk$ . The equilibrium level  $\dot{k} = 0$  is represented by an inverted-U-shaped curve in Fig. 9.2. As you can see from Fig. 9.2, per capita consumption increases with  $k$  up to certain level and declines afterwards. In the region below the curve,  $\dot{k} > 0$  (thus,  $k_{t+1} > k_t$ ). We have indicated the direction of such change in per capita capital stock by a right-facing blue arrow. If the economy is operating in the region above the curve,  $\dot{k} < 0$  (thus,  $k_{t+1} < k_t$ ). We have indicated the direction of such change in per capita capital stock by a left-facing blue arrow in Fig. 9.2.



**Fig. 9.2: Dynamics of  $k$** 

Now let us combine Fig. 9.1 and Fig. 9.2. Plotting all these lines and curves in the  $c$ - $k$  plane with  $c$  along the vertical axis and  $k$  along the horizontal axis, we get a diagram as shown in Fig. 9.3.


**Fig. 9.3: Phase Diagram**

The direction of arrows in the phase diagram shows the direction of movements of  $c$  and  $k$  when they are not on the  $\frac{dc}{dt} = 0$  and  $\frac{dk}{dt} = 0$  lines respectively. From Fig. 9.3 it is clear that there exists one steady state, denoted by point E in the diagram, which is associated with positive values of  $c$  and  $k$ . It can be shown (though we make no attempt to prove it here) that this steady state point is a ‘saddle point’ such that there exists a unique path which approaches the steady state point; all the other paths move away from the steady state point. This unique path (called saddle path) is denoted by the dotted purple line in the diagram. This is the only path which satisfies all the first order conditions of optimality (conditions (ia)-(iva)) including the transversality condition. This path is therefore the optimal path which satisfies the social planner’s dynamic optimization problem. Given any initial capital-labour ratio, the social planner will choose the initial per capita consumption so as to be on this optimal path, which will eventually take the economy to its long run equilibrium (or steady state) point E. If the economy is operating on any point on the saddle path, it will converge to the saddle point (E). If the economy, on the other hand, is operating at any other point on Fig. 9.3, it will move farther away from the steady state.

At this juncture it is important to point out certain important features of the steady state point E. As was mentioned before, at the steady state the per capita capital stock,  $k$ , and per capita consumption,  $c$ , are constant. Also, at this non-zero steady state, the  $k$ -value and the corresponding  $c$ -value are defined respectively by the equations:  $f'(k) = n + \rho$  and  $c = f(k) - nk$ . The condition that  $f'(k) = n + \rho$  is called the **modified golden rule**. It states that the steady state value of the capital-labour ratio is such that the marginal product of capital is equal to the sum of the rate of population growth and the rate of time

preference. In any dynamic model of capital accumulation and growth the concept of golden rule plays an important role. The **golden rule** itself refers to that steady state value of capital-labour ratio where  $f'(k) = n$ , i.e., the marginal product of capital is exactly equal to the rate of growth of population. The significance of the golden rule obtains from the fact that it is that steady state value of capital-labour ratio which corresponds to the maximum steady state value of per capita consumption. Note that when households' rate of time preference is zero, the modified golden rule coincides with the golden rule and the steady state of the Ramsey-Cass-Koopmans model is attained at the golden rule point. However, when people have positive rate of time preference, they are unwilling to sacrifice current consumption for future consumption beyond a point; as a result they accumulate less and reach a lower level of steady state consumption than in the golden rule.

### 9.3 DECENTRALIZED HOUSEHOLDS' PROBLEM

The Ramsey model described above was transformed by Cass and Koopmans into the problem of a decentralized competitive market economy consisting of many households – all with identical preferences. We shall see that under certain assumptions the optimal solution path in the two cases turns out to be identical.

Instead of assuming the existence of a central planner, let us consider a market economy where the households are price takers. There are two competitive factor markets which determine the wage rate ( $w_t$ ) and the interest rate or the rate of return on capital ( $r_t$ ) at each point of time.

There exist many identical competitive firms which hire in labour and capital from the households at the above mentioned wage rate and rental rate. Using labour and capital these firms produce the final output, using the same technology as the central planner. Competition ensures that the wage rate and the rental rate are equal to the respective marginal products of labour and capital at full employment; i.e.,  $r_t = f'(k_t)$  and  $w_t = f(k_t) - k_t f'(k_t)$ .

Each household maximizes its welfare given as before. Its objective function is to

maximize  $W = \int_0^{\infty} u(c_t) \exp^{-\rho t} dt$ . Each household starts with a certain amount of

capital stock and labour stock. Additionally, households can borrow from one another. Let  $a_t \equiv k_t - b_t$  denote the per capita asset stock of an household at period  $t$ , which consists of the per capita capital stock owned by the household at period  $t$  minus the amount of debt (per capita) at period  $t$ . (If the household is a net lender, then  $b_t$  would be negative). Arbitrage condition in the asset market ensures that physical capital and lending earns the same rate of return. Hence the per capita income of the household at period  $t$  is  $w_t + r_t a_t$ , which the household spends on consumption and further asset accumulation. Thus the budget

constraint faced by the household in per capita terms is given by:

$$c_t + \frac{da}{dt} + na_t = w_t + r_t a_t.$$

The complete dynamic optimization problem for the household can now be written in terms of the two time dependent variables: per capita consumption ( $c_t$ ) and per capita asset stock ( $a_t$ ). Our optimization problem is as follows:

$$\text{Maximize } W = \int_0^{\infty} u(c_t) \exp^{-\rho t} dt$$

subject to

$$\frac{da}{dt} = w_t + (r_t - n)a_t - c_t$$

In the above problem,  $a_0$  is given exogenously.

The corresponding Hamiltonian function and the first order conditions in terms of the control variable ( $c_t$ ), state variable ( $a_t$ ) and co-state variable ( $\lambda_t$ ) are as follows:

$$H = u(c_t) \exp^{-\rho t} + \lambda_t [w_t + (r_t - n)a_t - c_t]$$

$$(ib) \quad u'(c_t) \exp^{-\rho t} = \lambda_t$$

$$(iib) \quad \frac{d\lambda}{dt} = -\lambda_t(r_t - n)$$

$$(iiib) \quad \frac{da}{dt} = w_t + (r_t - n)a_t - c_t$$

$$(ivb) \quad \lim_{t \rightarrow \infty} \lambda_t a_t = 0$$

Note that these conditions look quite similar to the first order conditions that we obtained for the central planner's problem (conditions (ia)-(iva) in the previous section). However, for the household's problem there is an additional condition which has to be satisfied if we want to get a meaningful solution. This condition is called the **No-Ponzi-Game condition** which we elaborate below. First note that we have allowed for intra-household borrowing which means that a household can maintain a consumption stream above its income by borrowing. Now if a household could go on borrowing indefinitely then it will be optimal for the household to always maintain an infinitely high (or maximum possible) consumption stream and finance such high level of consumption simply by borrowing more and more. Of course, such a strategy would also imply that the net per capita borrowing of the household would increase exponentially at the rate  $(r_t - n)$  (since the household has to borrow not only for consumption, but also to pay back the interest rate as well) and the present discounted value of the net debt of the household will approach infinity. Such a financing scheme is called Ponzi-Game financing. In order to rule out such infinite indebtedness by any family we specify the condition that the even though the household can be

temporarily a net debtor (i.e., the present discounted value of its asset is temporarily negative), over a sufficiently longer horizon, it must eventually repay all its debt and hold non-negative asset stock. Formally, as  $t$  goes to infinity, the present discounted value of the (per capita) asset stock of the household must be non-negative:

$$(vb) \lim_{t \rightarrow 0} a_t \exp^{-\int_0^t (r_t - n) d\tau} \geq 0.$$

Condition (vb) is called the No-Ponzi-Game condition. For a decentralized household conditions (ib)-(vb) specify the optimal path that solves the household's dynamic optimization problem.

As before, differentiating (ib) with respect to  $t$  and using (iib) to eliminate  $\lambda_t$ , and noting that the term  $\frac{-cu''(c)}{u'(c)} \equiv \sigma$  denotes the elasticity of marginal utility with respect to consumption, we can derive the following differential equation:

$$(vib) \frac{dc}{dt} = \left( \frac{c_t}{\sigma} \right) (r_t - n - \rho)$$

Equations (iiib) and (vib) together form a system of differential equations which, along with the transversality condition and the no-Ponzi-Game condition, determine the movements of per capita consumption and per capita asset stock of the household along the optimal path.

Note that while for the economy as a whole the wage rate and the rental rate are given by  $r_t = f'(k_t)$  and  $w_t = f(k_t) - k_t f'(k_t)$ , the price-taking households take these as given while optimizing. However, we assume that the households are endowed with perfect foresight. It implies that households **do not know** the relationship between  $w$  and  $r$  with the per capita capital stock; but they can exactly **guess** the values of  $w$  and  $r$  at every point of time.

Also note that while the households can borrow from one another, in equilibrium for the economy as a whole, net borrowing must be zero. Since households are all identical, this implies that net borrowing of each household would be zero, i.e.,  $a_t = k_t$ .

Thus, replacing  $r_t = f'(k_t)$ ,  $w_t = f(k_t) - k_t f'(k_t)$ , and  $a_t = k_t$  in equations (iiib) and (vib), we get a system of differential equations which is exactly identical to the system of differential equations that we obtained for the central planner. Hence, their solution must also be the same. In other words, under the assumption of perfect foresight on the part of the households, *the household's optimal consumption and accumulation paths are exactly identical to the consumption and accumulation paths that would be chosen by a social planner*. To put it differently, the decentralized market economy's solution path coincides with the solution path of the centralized command economy, and the solution paths can once again characterized by the diagram given earlier. This is a very

strong result which shows the equivalence between the market equilibrium and the socially optimal solution. This equivalence result, however, depends crucially on the assumption of perfect foresight.

### Check Your Progress 1

1. What are the steady state values of per capita consumption and per capita capital stock in the centralized version of the Ramsey optimal growth model?

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2. Would the above values change if we consider a decentralized economy instead of a centralized planner's problem? Why?

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## 9.4 GOVERNMENT IN THE RAMSEY-CASS-KOOPMANS MODEL: RICARDIAN EQUIVALENCE

The decentralized equilibrium that we discussed above does not incorporate government as a separate economic agent. It is however easy to introduce a role for the government in the de-centralized Ramsey-Cass-Koopmans model. There are two options to finance government spending: (i) taxation, and (ii) borrowing. Suppose the government finances its expenditure by imposing a lump-sum tax on the household's income in every period. Since the government expenditure is unrelated to the households' consumption expenditure, the utility function of the households remains unchanged. However, since the government is taking away a part of the household's income, the budget constraint of the household changes. To be more precise, if the government imposes a tax  $\tau$  on the household income then the disposable income of the household reduces to  $w_t + r_t a_t - \tau$ . Consequently, the budget constraint of the household becomes:  $\frac{da}{dt} = w_t + (r_t - n)a_t - c_t - \tau$ . The rest of the optimal conditions, discussed in Section 9.3, remains the same.

In the phase diagram therefore, the  $\frac{dk}{dt} = 0$  curve shifts downward by the amount  $\tau$ , as shown in Fig. 9.4. The new steady state is obtained at the point E' in Fig. 9.4, where the steady state value of the per capita capital stock remains the same, but the steady state value of per capita consumption is lower. The corresponding optimal path of the household is once again denoted by the dotted line in the diagram.

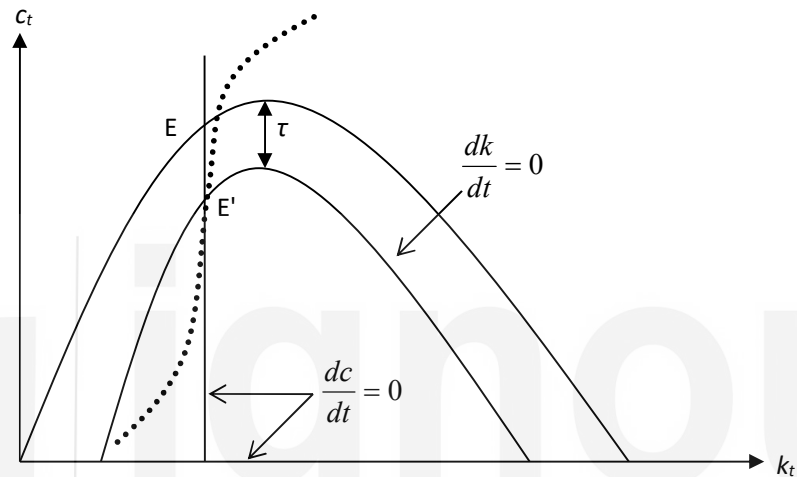


Fig. 9.4: Phase Diagram (with Government)

Now suppose the government decides to finance its expenditure by borrowing from the households, instead of taxation. Will that make any difference to the households' optimal condition? At first glance it seems that if the government borrows from the households by distributing government bonds, then the government bonds would constitute an asset for the households (because the government would pay an interest on these loans). So the household's disposable income is not reduced; only the composition of its asset portfolio changes. So at first glance it seems that the household's optima under tax financing and under debt financing would be different. However, a more careful consideration of the problem reveals that the mode of financing of the government expenditure really **does not** make any difference to the household's optima. As long as the government is forced to maintain a balanced budget in the long run (i.e., it cannot go on borrowing indefinitely), at some point of time in future the government has to impose a tax in order to repay its debt incurred earlier. Since the households have perfect foresight, they would know that at some point of time they would have to pay the tax. Thus, even though households are not paying the taxes today, they will adjust their current consumption and capital accumulation decisions accordingly which will enable them to bear the tax burden in future. This is a strong result: It states that for households endowed with perfect foresight (or rational expectation), the mode of financing of government expenditure does not matter. Tax financing and debt financing are exactly equivalent as far as the

household optima is concerned. This result is known as the *Ricardian Equivalence*.

### Check Your Progress 2

1. In the decentralized version of the optimal growth model, how would the equilibrium values of per capita capital stock and the per capita consumption change if the government imposes a lump sum tax to finance its own expenditure? In this context, explain the Ricardian Equivalence proposition.

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## 9.5 LET US SUM UP

In this unit, we have looked at the Ramsey-Cass-Koopmans optimal growth model where infinitely-lived individuals maximize their life-time utility. We identify the optimal consumption-accumulation path as the saddle path which takes the economy to its steady state in the long run. The steady state point is identified with the modified golden rule point. Although the problem is identified as a social planner's problem, its basic conclusion remains unchanged even when the economy is characterized by competitive decentralized households. Moreover introducing a government sector which finances its expenditure through lump sum taxation does not have any effect on the steady state values of the model.

## 9.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

### Check Your Progress Exercises 1

1. See Fig. 9.3. The steady state values of per capita consumption and per capita capital stock is given by the intersection of the equations  $f'(k) = n + \rho$  and  $c = f(k) - nk$ .
2. The steady state values of per capita consumption and per capita capital stock are the same in both central planner and decentralized household problems

### Check Your Progress Exercises 2

1. If the government imposes a lump sum tax, the steady state value of per capita capital stock remains unchanged while the per capita consumption would be lower. See Section 9.4 for details.

## 9.7 MATHEMATICAL APPENDIX

### Dynamic Optimization in Continuous Time (Optimal Control)

#### A.1 Finite Horizon Problem

Consider the following optimization problem which is defined over a finite time horizon from 0 to  $T$ :

$$\begin{aligned} \text{Max. } W &= \int_0^T F(u_t, x_t, t) dt, \text{ subject to } \frac{dx}{dt} = f(u_t, x_t, t) \\ x_0 &= \bar{x}; \quad u_t \in U. \end{aligned} \quad \dots(1)$$

The objective function here is an *integral*, and our task is to find out a time path of the time dependent variable  $u$  from the choice set  $U$ , (i.e., to choose a  $u \in U$  for each point of time  $t$  starting from 0 to  $T$ ) such that this integral is maximized.

But our choice is not unconstrained. (Had it been so, a simple point-by-point static optimization exercise would have given us the required solution path). Note that the  $F$  function (which can be called the *instantaneous* objective function) depends not only on  $u$  but also on another time dependent variable  $x$ . And our choice of  $u$  at each point of time affects the next period's value of  $x$  through the given differential equation. Thus our choice of  $u$  affects the instantaneous objective function directly, as well as indirectly through  $x$ . So when we are choosing the optimal time path for  $u$  that maximizes the integral  $W$ , we have to take into account how  $x$  is changing due to our choice of  $u$  – this explains the presence of a constraint function in the form of a differential equation in  $x$ .

Since we are choosing the variable  $u$  directly (i.e., we have direct control over this variable) it is called the *control variable*. The other time dependant variable,  $x$ , which changes over time due to our choice of  $u$ , but we don't have direct control over its change, is called the *state variable*. The function  $f$  in the given differential equation that describes the change in  $x$  over time is called the *state transition function*.

**Pontryagin's Maximum Principle:** Let  $u_t^*$  be a solution path to the problem specified in (1), and let  $x_t^*$  be the associated path for the state variable, where  $u_t^*$  is a piece-wise continuous function of  $t$  and  $x_t^*$  is a strictly continuous but piece-wise differentiable in  $t$ . Then there exist a strictly continuous and piece-wise differentiable variable  $\lambda_t$ , and a function  $H$  defined as:

$$H(u, x, \lambda, t) = F(u_t, x_t, t) + \lambda_t f(u_t, x_t, t),$$

such that

$$(i) \quad H \text{ is maximized with respect to } u \text{ at } u^* \text{ for all } t \in [0, T]$$

$$(ii) \quad \left. \frac{\partial H}{\partial x} \right|_{(u^*, x^*, \lambda, t)} = -\frac{d\lambda}{dt}$$

$$(iii) \quad \frac{\partial H}{\partial \lambda} \Big|_{(u^*, x^*, \lambda, t)} = \frac{dx}{dt}$$

$$(iv) \quad \lambda_T x_T = 0$$

The function  $H$  is called the **Hamiltonian Function** associated with the given dynamic optimizations problem, and the newly introduced time dependent variable  $\lambda$  is called the **co-state variable** associated with the state variable  $x$ .

The co-state variable  $\lambda_t$  measures the change in the value of the objective function  $W$  associated with an infinitesimal change in the state variable  $x$  at time  $t$ . If there were an exogenous tiny increment to the state variable at time  $t$ , and if the problem were modified optimally there after, then the increment in the total value of the objective would be  $\lambda_t$ . Thus it is the marginal valuation of the state variable at time  $t$ , and is therefore often referred to as the **shadow price** of the state variable at time  $t$ .

Pontryagin's Maximum Principle gives us four **first order necessary conditions** for the optimization problem defined in (1). The first three first order necessary conditions (F.O.N.C) are defined in terms of the Hamiltonian function. Note that if the Hamiltonian function is non-linear in  $u$ , then (i) can be replaced by the condition  $\frac{\partial H}{\partial u} \Big|_{(u^*, x, \lambda, t)} = 0$ , **provided the second order check**  $\frac{\partial^2 H}{\partial u^2} \Big|_{(u^*, x, \lambda, t)} < 0$

is verified.

The last condition of the Maximum Principle (condition (iv)), which specifies a terminal condition for  $\lambda$ , is called the **Transversality Condition**.

## A.2 Infinite Horizon Problems

Consider the following infinite horizon dynamic optimization exercise:

$$\begin{aligned} \text{Max. } W &= \int_0^{\infty} F(u_t, x_t, t) dt, \text{ subject to } \frac{dx}{dt} = f(u_t, x_t, t) \quad \dots(2) \\ x_0 &= \bar{x}; \quad u_t \in U. \end{aligned}$$

This problem is almost the same as the problem specified in (1); only now the time horizon is no longer finite.

When the time horizon stretches from 0 to  $\infty$ , additional complications may arise: First, the integral  $W$  may not converge. In that case it does not make sense to talk about optimal paths, because any paths  $(u_t, x_t)$  will eventually make the integral infinitely large, and no comparison between one set of  $(u_t, x_t)$  with another is possible. Hence, for the dynamic optimization exercise to be meaningful, we must impose additional restrictions so that the integral converges to a finite value.

Second, there is some controversy regarding the appropriate transversality condition in infinite horizon problems. Usually some limiting conditions, which

are similar to the transversality conditions applied in the finite horizon problems, are applied.

Consider problem (2) and suppose the integral converges. The F.O.N.C.s for this problem are given in the following theorem:

Let  $u_t^*$  be a solution path to the problem specified in (5), and let  $x_t^*$  be the associated path for the state variable, where  $u_t^*$  is a piece-wise continuous function of  $t$  and  $x_t^*$  is a strictly continuous but piece-wise differentiable in  $t$ . Then there exist a strictly continuous and piece-wise differentiable variable  $\lambda_t$ , and a function  $H$  defined as:

$$H(u, x, \lambda, t) = F(u_t, x_t, t) + \lambda_t f(u_t, x_t, t),$$

such that

(i)  $H$  is maximized with respect to  $u$  at  $u^*$  for all  $t \in [0, \infty)$

$$(ii) \left. \frac{\partial H}{\partial x} \right|_{(u^*, x^*, \lambda, t)} = - \frac{d\lambda}{dt}$$

$$(iii) \left. \frac{\partial H}{\partial \lambda} \right|_{(u^*, x^*, \lambda, t)} = \frac{dx}{dt}$$

$$(iv) \lim_{t \rightarrow \infty} \lambda_t x_t = 0$$

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# UNIT 10 OVERLAPPING GENERATIONS MODEL\*

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## Structure

- 10.0 Objectives
- 10.1 Introduction
- 10.2 Structure of the Model
- 10.3 Dynamic Inefficiency in Overlapping Generations Model
- 10.4 Social Security
- 10.5 Let Us Sum Up
- 10.6 Answer/Hints to Check Your Progress Exercises

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## 10.0 OBJECTIVES

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After going through this Unit you should be able to explain

- the standard two-period overlapping generations model with production;
- the concept of dynamic efficiency and examine whether the dynamic efficiency property holds for an overlapping generations economy; and
- the role of social security system in the overlapping generations framework in eliminating dynamic inefficiency.

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## 10.1 INTRODUCTION

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In the context of intertemporal decision making on the part of the households, you have come across the optimal growth model in Unit 9. There is another important framework that also considers households' intertemporal decision making – although over finite time horizon. This framework is called the overlapping generations framework. The framework was first developed by Samuelson (1954) which has subsequently been widely used in macro dynamics.

In the overlapping generations framework, the individuals have a finite time horizon (in the standard case, a two-period time horizon), but the society lives forever. The term 'overlapping generations' implies that at each point of time the life-time of two generations overlaps. To clarify, let us take the standard case where each generation lives exactly for two periods. Thus, for the cohort of individuals who are born at the beginning of period ' $t$ ', they are alive in period ' $t$ ' (when they are young) and in period  $(t + 1)$  (when they are old). On the other hand, a new set of individuals are born at the beginning of period  $(t + 1)$ , who would be alive in period  $(t + 1)$  and period  $(t + 2)$ . Thus, between the lifetimes of these two successive generations, there is an exact overlap of one period. In the following discussion we shall concentrate on the two-period overlapping generations framework only.

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## 10.2 STRUCTURE OF THE MODEL

The set of people who are born at the beginning of period  $t$  will be called 'generation  $t$ '. Let us now look at the activities of a representative member of generation  $t$ . Each person is born with an endowment of one unit of labor. In the first period of his life, when he is young, he works and earns a wage income of which he consumes a part and saves (and invests) the rest. In the next period of his life, when he is old, he does not work anymore. He nonetheless earns an interest income on his previous period's savings (investment). He also gets back the principal amount that he invested, which he consumes in the second period along with the interest earning. Thus, his first and second period budget constraints are respectively given by:

$$(i) \quad c_{1t} + s_t = w_t ;$$

$$(ii) \quad c_{2t} = (1 + r_{t+1})s_t ,$$

where  $c_{1t}$  and  $c_{2t}$  are the first and second period consumption of the representative member of generation  $t$ . In the above equations,  $w_t$  and  $r_{t+1}$  are the wage rate at period  $t$  and the rate of interest at period  $(t + 1)$  respectively.<sup>1</sup> The representative member maximizes his two-period utility function  $U(c_{1t}, c_{2t})$

subject to these two budget constraints. Noting that  $s_t = \frac{c_{2t}}{(1 + r_{t+1})}$ , we can combine the two budget constraint by eliminating  $s_t$ , which gives us the following single equation that represents the **life-time budget constraint** of the agent:  $c_{1t} + \frac{c_{2t}}{(1 + r_{t+1})} = w_t$ . Maximizing  $U(c_{1t}, c_{2t})$  subject to the life-time budget constraint generates the following two first order conditions:

$$(1) \quad \frac{U_1}{U_2} = (1 + r_{t+1}) ,$$

$$(2) \quad c_{1t} + \frac{c_{2t}}{(1 + r_{t+1})} = w_t .$$

From these two equations we can write the optimal consumption levels  $c_{1t}$  and  $c_{2t}$  as functions of  $w_t$  and  $r_{t+1}$ . Since,  $s_t = \frac{c_{2t}}{(1 + r_{t+1})}$ , the optimal value of  $s_t$  also becomes a function of  $w_t$  and  $r_{t+1}$ .

We shall assume that all members of all generations are identical in terms of tastes and preferences, i.e., they have **similar** utility functions.<sup>2</sup>

<sup>1</sup> Note that though the member of generation  $t$  made his saving and investment decision at period  $t$ , he earns the interest on that savings only in the next period. Hence  $r_{t+1}$  is the relevant rate of interest.

<sup>2</sup> Similar, but not identical. To be more precise, the time subscripts in the utility function will be different for different generations.

Let us now look at the overall macroeconomic picture. As we have mentioned before, at each period there are two generations who are simultaneously alive. Thus, at period  $t$ , there is a set of people who belong to generation  $(t - 1)$  (these are the people who are currently old) and a set of people who belong to generation  $t$  (these are the people who are currently young). The generation  $t$  people are the workers in period  $t$ , each of whom earn a wage income  $w_t$ . The individuals pertaining to generation  $(t - 1)$  are the interest earners; they earn an interest income on their previous period's savings (i.e., savings made in period  $(t - 1)$  at the rate  $r_t$ ).

Suppose, the population in successive generations grows at a constant rate  $n$ . Thus, if  $L_t$  denotes the number of people belonging to generation  $t$ , and  $L_{t-1}$  denotes the number of people belonging to generation  $(t - 1)$ , then  $L_t = (1 + n)L_{t-1}$ . Total population in the economy at period  $t$  is given by  $(L_t + L_{t-1})$ .

The production side of the economy is like any standard neoclassical growth model that you have seen before. A single final commodity is produced which is used both as a consumption good as well as an investment good. Technology for final commodity production is given by a neoclassical production function:  $Y_t = F(K_t, L_t)$ , where  $F$  is continuous, concave, exhibits constant returns to scale (CRS) with respect to its two factors – capital ( $K$ ) and labour ( $L$ ). As in the Ramsey model, the CRS property of the production function implies that per capita output ( $y$ ) can be written as the function of the capital-labour ratio ( $k$ ) in the following way:

$$y \equiv \frac{Y}{L} = \frac{F(K, L)}{L} = \frac{L F(K/L, 1)}{L} = F\left(\frac{K}{L}, 1\right) \equiv f(k).$$

Moreover the marginal products of capital and labour can also be written as the following functions of the capital-labour ratio:

$$\frac{\partial F}{\partial K} = f'(k) ; \quad \frac{\partial F}{\partial L} = f(k) - kf'(k).$$

The continuity and concavity properties of  $F(L, K)$  ensure that  $f(k)$  is also continuous and concave. Additionally we assume that the 'Inada Conditions'<sup>3</sup> hold:  $f(0) = 0$ ;  $f'(0) = \infty$ ;  $f'(\infty) = 0$ .

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<sup>3</sup> Named after Japanese economist Ken-Ichi Inada, these conditions refer to the shape of a production function or utility function. For the intensive form of the production function, the Inada conditions are as follows:

- (a)  $\lim_{k \rightarrow 0} f'(k) = \infty$ . This implies that when capital stock is small  $MP_K$  is very large.
- (b)  $\lim_{k \rightarrow \infty} f'(k) = 0$ . The  $MP_K$  is very small when capital stock is too large.

In a market economy with perfect competition, the wage rate and the rate of interest are equated with the respective marginal products of labour and capital. Thus  $w_t = f(k_t) - k_t f'(k_t)$  and  $r_t = f'(k_t)$ .

For the economy as a whole, savings-investment equality (which generates goods market equilibrium for the aggregate economy) tells us that  $C_t + I_t = Y_t = F(K_t, L_t)$ . Noting that investment is nothing but the augmentation of the capital stock (assuming no depreciation), i.e.,  $I_t = K_{t+1} - K_t$ , and also noting that due to the property of CRS  $F(L_t, K_t) = w_t L_t + r_t K_t$ , we can write the above savings-investment equality condition as:  $C_t + K_{t+1} - K_t = w_t L_t + r_t K_t$ . Notice that  $C_t$  denotes the aggregate consumption at g period  $t$ , which includes consumption by the old group and consumption by the young group. There are  $L_{t-1}$  number of old people who are alive at period  $t$  and each of them consume an amount  $c_{2t-1}$ .<sup>4</sup> On the other hand, there are  $L_t$  number of people at period  $t$  belonging to the young generation, each of them consuming an amount  $c_{1t}$ . Hence  $C_t = L_{t-1}c_{2t-1} + L_t c_{1t}$ .

At this point it is important to recall that each young person is born with an endowment of one unit of labour. There is no bequest; therefore the young people do not own any capital stock at period  $t$  (capital stock being the only asset in this economy). Thus, the entire capital stock is owned by the older generation. And since the older generation is going to die at the end of this period, they consume their entire interest earning plus the capital stock (which, in the one good world, is directly consumable). Hence,  $L_{t-1}c_{2t-1} = K_t + r_t K_t$ . Again, each young person earns a wage  $w_t$ , of which he consumes  $c_{1t}$  and saves  $s_t$ . Thus,  $L_t c_{1t} = L_t(w_t - s_t)$ . Using all these information, we can write the savings-investment equality condition as:

$$K_t + r_t K_t + L_t(w_t - s_t) + K_{t+1} - K_t = w_t L_t + r_t K_t.$$

On simplification of the above equation, we obtain the goods market equilibrium condition for the aggregate economy as:  $K_{t+1} = L_t s_t$ . Recall that  $s_t$  is a function of  $w_t$  and  $r_{t+1}$ . Further,  $w_t$  and  $r_{t+1}$  are functions of  $k_t$  and  $k_{t+1}$  respectively. Thus,  $s_t$  can be written as a function of  $k_t$  and  $k_{t+1}$ :  $s(w(k_t), r(k_{t+1}))$ . This allows us to write the goods market equilibrium condition in terms of the capital labour ratio as follows:

$$\begin{aligned} K_{t+1} &= L_t s(w(k_t), r(k_{t+1})) \\ \Rightarrow \frac{K_{t+1}}{L_{t+1}} &= \frac{L_t s(k_t, k_{t+1})}{L_{t+1}} = \frac{L_t s(w(k_t), r(k_{t+1}))}{(1+n)L_t} \\ \Rightarrow k_{t+1} &= \frac{s(w(k_t), r(k_{t+1}))}{(1+n)}. \end{aligned}$$

<sup>4</sup>  $c_{2t-1}$  is the 2nd period consumption of the representative member of generation  $(t-1)$ .

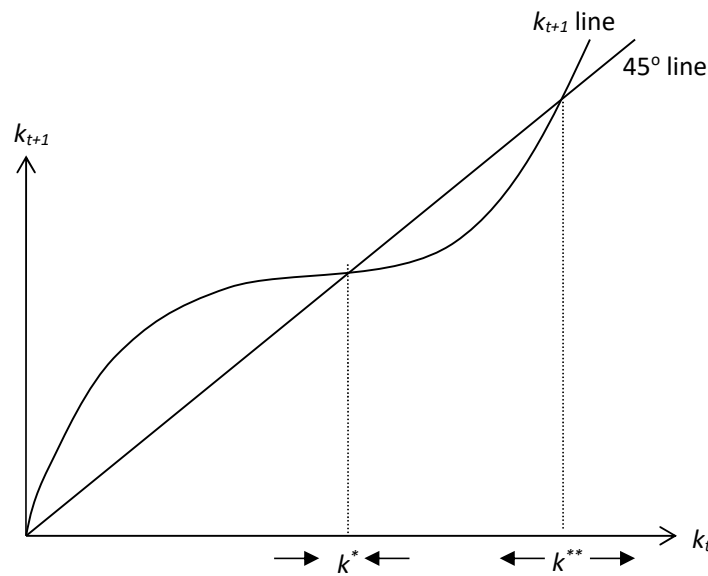
This last line above represents the basic dynamic equation of this overlapping generations model which specifies the relationship between the capital-labour ratio of today and the capital-labour ratio of tomorrow. Tracing this dynamic equation will tell us how the capital-labour ratio of the economy changes over time. It is easy to see that this equation is a first order non-linear difference equation in  $k$ . To characterize the solution path we shall use the phase diagram technique. The phase diagram plots  $k_{t+1}$  on the vertical axis and  $k_t$  on the horizontal axis. The intersection of the  $k_{t+1}$  line with the  $45^\circ$  line denotes the steady state. Let us now determine the slope of this line. The slope is given by the derivative  $\frac{dk_{t+1}}{dk_t}$ . As you can see, the LHS of the dynamic equation is also a function of  $k_{t+1}$ . Hence, after total differentiation of both the sides, we obtain

$$dk_{t+1} = \frac{s_w \frac{dw_t}{dk_t} dk_t + s_r \frac{dr_{t+1}}{dk_{t+1}} dk_{t+1}}{(1+n)} .$$

Noting that  $\frac{dw}{dk} = -kf''(k)$  and  $\frac{dr}{dk} = f''(k)$  we can write the above equation as:

$$\frac{dk_{t+1}}{dk_t} = \frac{s_w [-k_t f''(k_t)]}{(1+n) - s_r f''(k_{t+1})} .$$

Notice that as yet we do not know the signs of  $s_w$  and  $s_r$ ; hence we cannot say whether the  $k_{t+1}$  line is positively sloping or negatively sloping. It is easy to see that under the assumption that consumption in both periods are normal goods (i.e., both  $c_1$  and  $c_2$  increases with an increase in  $w$ ),  $0 < s_w < 1$ . The sign of  $s_r$  however is ambiguous. Since an increase in  $r$  implies that the relative price of future consumption in terms of current consumption falls. Hence, due to substitution effect current consumption should fall, which means that with unchanged wage rate, savings would rise. However, a fall in relative price will also be associated with a positive income effect on current consumption. Thus, whether current consumption increases due to an increase in  $r$  depends crucially on whether the income effect dominates the substitution effect. If the income effect dominates the substitution effect, then  $c_1$  rises and consequently  $s_r < 0$ . On the other hand, if the substitution effect dominates the income effect, then  $c_1$  falls and consequently  $s_r > 0$ . We shall assume here that the latter holds, i.e.,  $s_r > 0$ . Under this assumption the  $k_{t+1}$  line is positively sloping. We still do not know the curvature of this line. Depending on the curvature, multiple equilibria (i.e., multiple steady states) are possible, as shown in Fig. 10.1.



**Fig. 10.1: Multiple Steady State**

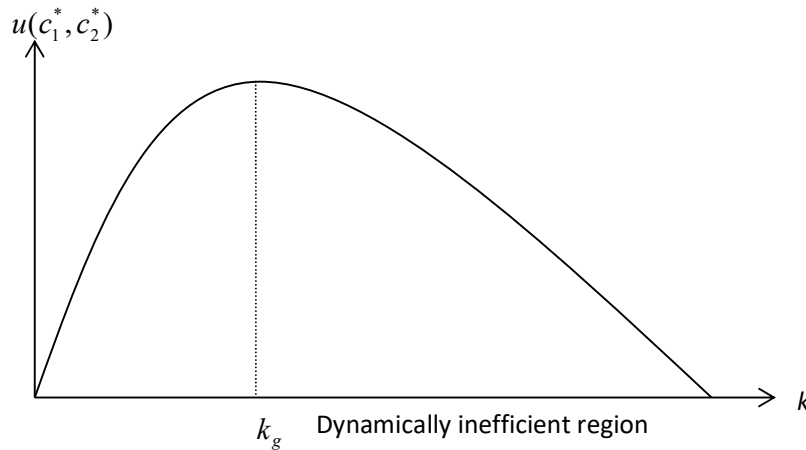
Local stability of a steady state depends on whether the  $k_{t+1}$  line intersects the 45° line from above or from below. In the above diagram  $k^*$ , where the  $k_{t+1}$  line intersects the 45° line from above, is a locally stable equilibrium. On the other hand,  $k^{**}$ , where the  $k_{t+1}$  line intersects the 45° line from below, is a locally unstable equilibrium.

### 10.3 DYNAMIC INEFFICIENCY

In the following discussion we shall assume that the parametric conditions are such that there exists a unique equilibrium which is locally stable. Even when the steady state is unique and stable, in an overlapping generation framework the equilibrium can still be characterized by dynamic inefficiency. This is a serious shortcoming of the overlapping generations framework in comparison to the optimal growth framework. Before we elaborate this point further it is important to know what we mean by dynamic efficiency or inefficiency.

The concept of dynamic efficiency is closely related to the concept of Pareto efficiency. To understand the concept, let us consider a situation where we are comparing between various possible steady states or equilibrium points. Each of these steady states is characterized by a different capital-labour ratio and corresponding steady state levels of per capita consumption for the old and the young. Since utility depends on the consumption during youth and during old-age, each of these steady states is therefore associated with a different level of steady state utility. Now, among all these steady states, the one which provides the maximum steady state value of utility is called the ‘golden rule’ point and the corresponding capital-labour ratio is called the ‘**golden rule**’ capital labour ratio. This point is the best possible steady state point which provides maximum lifetime utility to each person. One can show that steady state utility is maximized at the point where  $f'(k) = n$ . Thus, golden rule capital-labour ratio is defined by

$k_g$  such that  $f'(k_g) = n$ . Fig. 10.2 depicts the golden rule capital labour ratio as the point which maximizes steady state utility.



**Fig. 10.2: Golden Rule Capital Labour Ratio**

Now, consider all the points which lie to the right of  $k_g$ . Here, the capital-labour ratio is more than optimal. In other words, people are over-saving and over-investing. If instead of saving, people consume a part of their savings in the first period of their life, then the capital-labour ratio will fall to  $k_g$  and at the same time their life time utility will increase. Thus, all the point lying to the right of  $k_g$  are Pareto inferior to  $k_g$ : one can improve the current consumption without reducing future consumption and thus improve total life-time utility. These points are called dynamically inefficient points. Now, consider all the points to the left of  $k_g$ . Here also the utility level is less than  $k_g$ ; hence by moving to  $k_g$  one can improve the steady state value of utility. However such move is not costless any more. If one wants to move from a point to the left of  $k_g$  to  $k_g$ , then he has to save more. In other words, he has to forgo some amount of current consumption. Thus, his current utility would *fall*, even though he would be better off in the future (once he reaches  $k_g$ ). Since such a move from the left to right involves a current utility loss, we cannot say for sure whether these points are better or worse than  $k_g$ . All these points are Pareto efficient or dynamically efficient.

We have seen that in the optimal growth framework (for example, Ramsey-Cass-Koopmans model) the steady state is defined by  $f'(k) = n + \rho$ . Thus, the steady state point in that framework is always to the left of the golden rule point and is therefore dynamically efficient. In the case of the overlapping generations framework however dynamic efficiency of the equilibrium point cannot be guaranteed. In fact under very reasonable parametric values the steady state could be dynamically inefficient.

To see how, let us consider the following example where we assume a specific utility function and a specific production function. Let the utility function of the representative member of generation  $t$  be  $U(c_{1t}, c_{2t}) \equiv \log c_{1t} + \log c_{2t}$ . Also let

the per worker production function be  $f(k_t) \equiv Ak_t^\alpha$ ,  $0 < \alpha < 1$ . Both the log utility function and the Cobb-Douglas production function are known to be very well-behaved which satisfy all the standard neoclassical properties.

With the log utility function as specified above, one can easily verify that the first order conditions for individual's utility maximization exercise are given by:

$$(i) \frac{c_{2t}}{c_{1t}} = (1 + r_{t+1})$$

$$(ii) c_{1t} + \frac{c_{2t}}{(1 + r_{t+1})} = w_t.$$

These two first order generates a savings function which is given by:  $s_t = \frac{1}{2} w_t$ .

Also, with Cobb-Douglas production function  $w_t = (1 - \alpha)Ak_t^\alpha$ . Thus, the basic dynamic equation in this example is given by:  $k_{t+1} = \frac{1}{(1+n)}(1 - \alpha)Ak_t^\alpha$ . At

steady state  $k_t = k_{t+1} = k^*$ . Hence putting  $k^*$  at the LHS and RHS of the above equation, we can solve for the steady state value as:  $k^* = \left[ \frac{A(1-\alpha)}{(1+n)} \right]^{1/(1-\alpha)}$ .

Let us now compare this steady state value of capital-labour ratio with the corresponding golden rule capital labour ratio. Note that with the Cobb-Douglas production function, the golden rule capital labour ratio in this example is defined

by:  $A\alpha(k)^{\alpha-1} = n$ . Solving, we get  $k_g$  as:  $k_g = \left[ \frac{A\alpha}{n} \right]^{1/(1-\alpha)}$ . Thus, whenever  $\frac{(1-\alpha)}{\alpha} > \frac{(1+n)}{n}$ ,  $k^* > k_g$ , i.e., the steady state will be dynamically inefficient.

For example, the equilibrium will be dynamically inefficient if  $\alpha = 1/4$  and  $n = 1$ .

An obvious question that arises here: Why is it that the steady state could be dynamically inefficient in the overlapping generations framework, while such possibility is ruled out in the optimal growth framework? The answer lies in the fact that in the overlapping generations individuals are selfish (no bequest). Since they do not have to share the benefits of their investment with their successive generations (who are growing at the rate  $n$ ), when they consider the possible future return to their investment, the return is **not** net of the population growth. To put it differently, the relevant return for them is not  $(f'(k) - n)$  but just  $f'(k)$ . Hence they would be interested in investing as long as this return is positive, even if it falls short of  $n$ . This is the reason for their possible over-saving.

### **Check Your Progress 1**

1. Derive the basic dynamic equation of the standard two period overlapping generations model with production. Explain the intuition behind this equation.

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2. What is dynamic efficiency? Is the steady state under the overlapping generations structure necessarily dynamically efficient? Give an example to elaborate your answer.

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## 10.4 SOCIAL SECURITY

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Given that the overlapping generations equilibrium could be characterized by dynamic inefficiency, we can ask if there is any role for the government here. In other words, does there exist any government policy which could check the over-saving by the households? Typically the government could influence the savings decision through some kind of social security programmes. What is a social security programme? A social security programme is meant to provide some income to individuals after retirement. There are usually two types of social security programmes which are in vogue – a fully funded system and a pay-as-you-go system.

In a fully funded system a certain stipulated amount is taken by the government from each young person at period  $t$  as social security contribution. The same amount is invested by the government and returned with interest to the old people at period  $(t + 1)$ . Since the young at period  $t$  are the old at period  $(t + 1)$ , in a fully funded system the set of people who made the contributions **today** and the set of people who will receive the corresponding benefit **tomorrow** are the same. If  $d_t$  is the contribution made by each young person at period  $t$ , and  $b_{t+1}$  is the benefit received by each old person at period  $(t + 1)$ , then  $b_{t+1} = (1 + r_{t+1})d_t$ .

In an unfunded pay-as-you-go system, a certain stipulated amount is taken by the government from each young person at period  $t$  and the entire amount is immediately distributed to the old people at period  $t$ . In other words, the pay-as-you-go system involves a transfer of funds from the current young to the current youth. Note that population is growing at the rate  $n$ ; therefore at each point of time there are  $n$  more people belonging to the currently young group compared to the currently old group. Thus, if  $d_t$  is the contribution made by each young

person at period  $t$ , and  $b_t$  is the benefit received by each old person at period  $t$ , then  $b_t = (1 + n)d_t$ .

The type of social security system has important implications for the savings decisions of the young. Note that in a fully funded system the government is doing part of the savings on behalf of the individuals. The representative individual is effectively saving an amount equal to  $(s_t + d_t)$  and in the next period earning an interest income equal to  $(1 + r_{t+1})(s_t + d_t)$ . Since each person knows that this is the amount that he would receive in the next period, in his own savings decision, the individual will optimally adjust (cut back) his own savings so that his total effective savings remains the same as in the pre-social security economy. Thus, a fully funded social security system has no impact on the total savings and capital accumulation. If the economy was in a dynamically inefficient steady state in the pre-social security economy, it will remain so even after introducing a fully funded social security system.

In contrast, in a pay-as-you-go social security system, the relevant rate of return on government securities for an individual is  $n$ . On the other hand, the rate of interest on capital accumulation is  $r$ . Thus, as long as  $r < n$ , it pays to save less and contribute more in the form of social security. Hence, if the economy was in a dynamically inefficient steady state in the pre-social security economy (which implies that  $r$  is indeed less than  $n$ ), introducing a pay-as-you-go type of social security system will reduce private savings and capital accumulation. As a result the economy will move to the dynamically efficient region.

### Check Your Progress 2

1. Does any type of social security system necessarily eliminate the dynamic inefficiency problem in the overlapping generations framework? Elaborate your answer with the two types of social security systems.

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## 10.5 LET US SUM UP

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In this unit, we have looked at the standard overlapping generations framework with production. We have derived the basic dynamic equation which basically states that the tomorrow's capita stock is equal to the savings by the young generations today. We have derived the steady state conditions. It has been shown that the steady state under the OLG framework may not be dynamically efficient. The government can play an important role here to ensure dynamic efficiency by introducing a social security system. However, the type of the social security system is important: a pay-as-you-go type of social security

system is effective in eliminating inefficiency while a fully funded social security system is not effective.

**Overlapping  
Generations Model**

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## **10.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES**

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### **Check Your Progress Exercises 1**

1. Read Section 10.2 and write your answer.
2. Read Section 10.3 and write your answer.

### **Check Your Progress Exercises 2**

1. The pay-as-you-go social security system is efficient when population is increasing. Read Section 10.4 and write your answer.

